

Ex.No.9 Use Process Explorer to identify suspicious processes

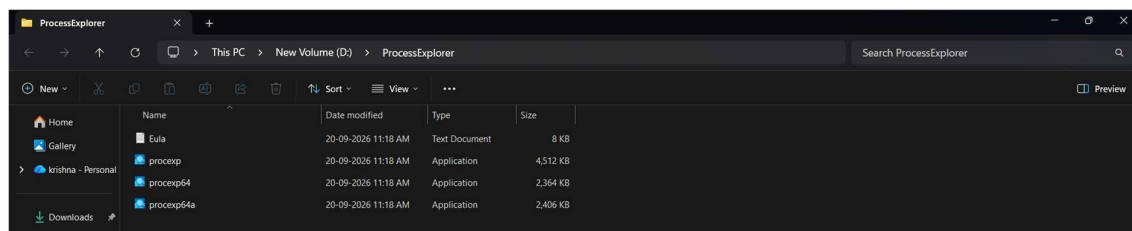
Date:

Description

Process Explorer is a powerful Windows tool that provides detailed information about running processes, allowing users to monitor, troubleshoot, and detect suspicious activities. It can be used to detect potential malware or harmful processes by analyzing various aspects of running applications. Here's a step-by-step guide on how to use Process Explorer to identify suspicious processes on your system:

Step 1: Download and Set Up Process Explorer

1. Download Process Explorer:
 - o Go to the official Microsoft Sysinternals website and download Process Explorer:
[Download Process Explorer](#).
2. Extract the Program:
 - o Once downloaded, extract the ZIP file to a folder.
3. Run Process Explorer:
 - o Open the folder where you extracted the files and run procexp64.exe (64-bit) or procexp.exe (32-bit) as Administrator by right-clicking and selecting "Run as Administrator."



Step 2: Familiarize Yourself with the Interface

When Process Explorer starts, you'll see an interface that lists all running processes, showing their hierarchy, memory usage, CPU usage, and much more.

1. Process Tree:
 - o The main window shows a tree of all running processes, with each child process listed under its parent.
 - o Processes are color-coded to indicate their status:
 - ☐ Pink: Suspended processes.
 - ☐ Light Blue: Processes running under the same user as you.
 - ☐ Dark Blue: Services or processes running under system accounts.
 - ☐ Green: New processes (appearing and disappearing quickly may be

suspicious).

- Red: Processes that have just exited.

2. Columns:

- o Various columns show information such as Process ID (PID), CPU usage, Memory usage, and more.

Step 3: Identify Suspicious Processes

Now that you are familiar with the interface, follow these steps to investigate potentially suspicious processes:

1. Look for Unfamiliar Processes:

- Scan the list of running processes and identify any names that you don't recognize. Some malware disguises itself with legitimate-looking names, but odd names can be a red flag.
- Legitimate processes should typically be from trusted sources (e.g., Microsoft, Intel, Adobe, etc.).

Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Secure System	172 K	61,404 K	156			
Registry	14,840 K	59,024 K	232			
System Idle Process	85.16	60 K	8 K	0		
System	0.74	48 K	3,568 K	4		
Interrupts	1.24	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe	1.156 K	1,416 K	876			
Memory Compression	< 0.01	1,748 K	6,333,000 K	3404		
csrss.exe	2,624 K	6,740 K	1172			
wininit.exe	1,696 K	8,908 K	1260			
services.exe	0.25	7,064 K	19,156 K	1404		
svchost.exe	< 0.01	12,956 K	40,996 K	1604	Host Process for Windows S...	Microsoft Corporation
WsmProvSE.exe		16,316 K	32,636 K	6572		
unsecapp.exe		1,836 K	9,508 K	6096		
WsmProvSE.exe		5,448 K	15,664 K	7500		
CrossDeviceService.exe	< 0.01	32,160 K	1,10,624 K	11860	Microsoft Cross Device Servi...	Microsoft Corporation
SearchIndexer.exe	0.12	73,920 K	1,32,064 K	12436		
msedgewebview2.exe		47,260 K	1,23,480 K	14176	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,960 K	12,604 K	13620	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		71,744 K	51,504 K	13144	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		13,696 K	39,656 K	13436	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		8,932 K	19,216 K	2072	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		71,544 K	86,648 K	14512	Microsoft Edge WebView2	Microsoft Corporation
WindowsExperienceHost.exe		1,12,500 K	1,68,444 K	15944	Windows Experience Host	Microsoft Corporation
PushNotificationsBackground.exe		2,148 K	17,908 K	12572	PushNotificationsBackground.exe	Microsoft Corporation
WidgBoard.exe	0.12	87,720 K	1,02,516 K	12648		
msedgewebview2.exe		45,600 K	1,27,080 K	4680	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		2,860 K	13,060 K	6396	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,04,848 K	95,476 K	12896	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		15,644 K	42,468 K	12900	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		13,900 K	23,540 K	6552	Microsoft Edge WebView2	Microsoft Corporation
msedgewebview2.exe		1,28,404 K	1,49,012 K	9616	Microsoft Edge WebView2	Microsoft Corporation
RuntimeBroker.exe		13,532 K	60,356 K	12756	Runtime Broker	Microsoft Corporation
UserDebugger.exe		2,012 K	11,816 K	13176	User Debugger	Microsoft Corporation
WidgService.exe		5,620 K	28,948 K	13300	WidgService.exe	Microsoft Corporation
ShellExperienceHost.exe	0.25	47,164 K	1,05,472 K	13004	Windows Shell Experience H...	Microsoft Corporation
RuntimeBroker.exe		2,304 K	13,696 K	14736	Runtime Broker	Microsoft Corporation
MicrosoftStardewProvi...		11,492 K	43,768 K	14008		
PhoneExperienceHost.exe		63,772 K	1,70,104 K	8188	Microsoft Phone Link	Microsoft Corporation
TaskScheduler.exe		1,33,872 K	1,31,112 K	15488		
RuntimeBroker.exe		3,500 K	24,000 K	16504	Runtime Broker	Microsoft Corporation
unsecapp.exe		1,728 K	10,056 K	18880	Sink to receive asynchronous...	Microsoft Corporation
RuntimeBroker.exe		1,620 K	10,612 K	1520	Runtime Broker	Microsoft Corporation
IGCC.exe		44,124 K	59,548 K	19508	IGCC	Intel Corporation
RuntimeBroker.exe		3,168 K	21,992 K	15260	Runtime Broker	Microsoft Corporation

2. Verify Digital Signatures:

- Right-click any suspicious process and select Properties.
- Go to the Image tab and check if the process has a valid Digital Signature.
- Legitimate applications from trusted developers will have valid signatures. Processes without signatures or with invalid signatures are suspicious.

3. Check the Path of the Process:

- In the Properties window of the suspicious process, look under the Image tab for the Path.
- Verify if the process is located in a legitimate directory. For example, legitimate system processes should reside in C:\Windows\System32.

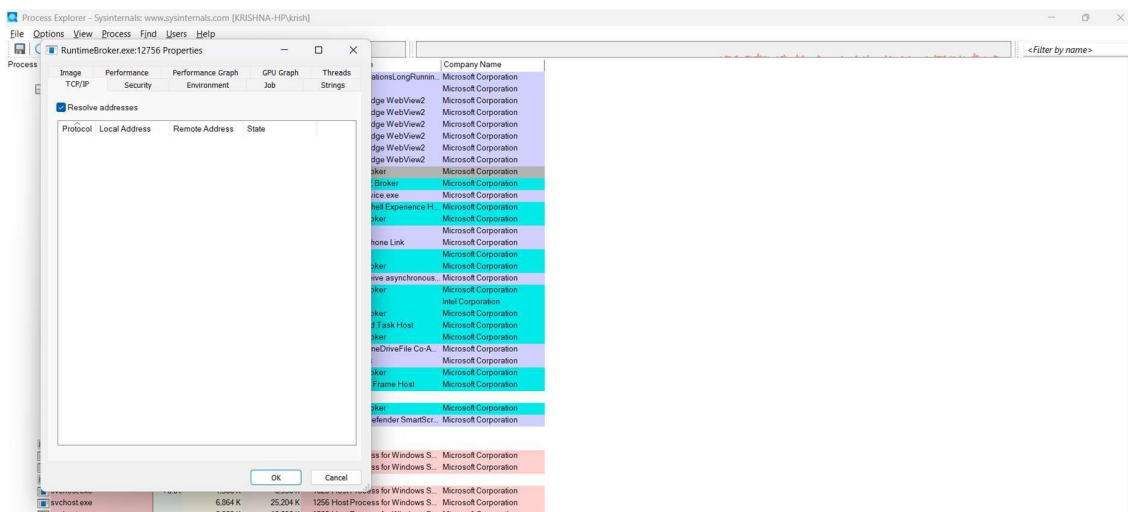
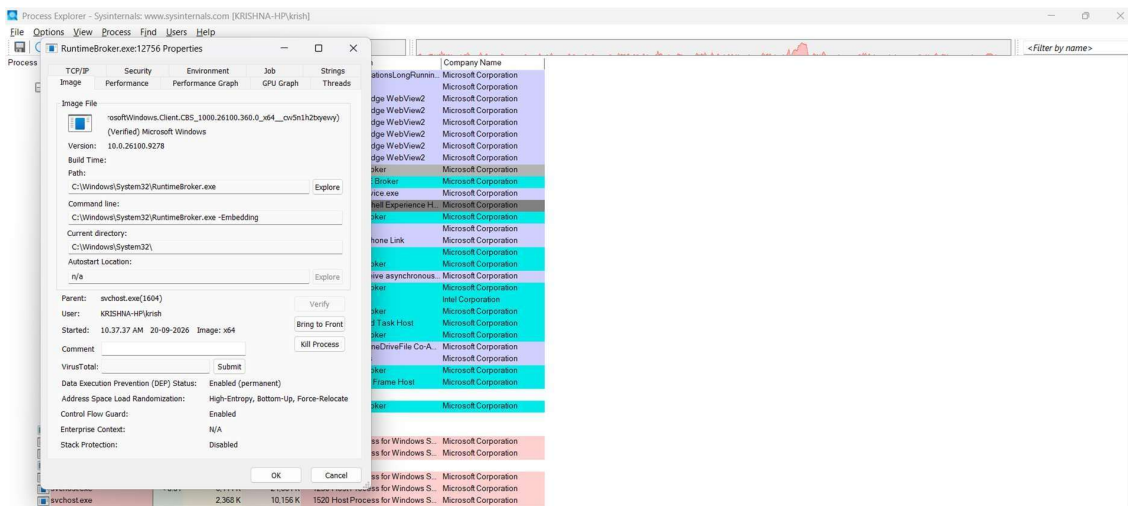
- If the process is running from a temporary folder, user folder, or a random directory, it might be malware.

4. Monitor CPU, Memory, and Disk Usage:

- Look at the CPU, Memory, and Disk usage columns.
- A process that is using an unusual amount of resources without explanation could be malicious. For example, high CPU usage on a process like svchost.exe can indicate malware.

5. Check Process Description and Company Name:

- Look at the Description and Company Name columns in Process Explorer.
- Legitimate processes usually provide clear descriptions and are tied to well-known companies.
- Lack of information or suspicious company names should prompt further investigation.



6. Check the Process Network Activity:

- Some malware communicates with external servers. You can inspect a process's network connections:

- o Right-click on the process and select Properties, then go to the TCP/IP tab to see network activity associated with the process.
- o Unexpected or unexplained external connections can indicate a malicious process.

Step 4: Perform a Google Search on Suspicious Processes If

you find a process that looks suspicious or unfamiliar:

1. Search online: Perform a quick Google search using the process name (e.g., processname.exe) to find out whether the process is known to be safe or malicious.
2. Check Malware Databases: Use online databases such as VirusTotal or ProcessLibrary to verify whether the process is known for being part of malware.

Step 5: Kill or Suspend Suspicious Processes If you

find a process that you believe is malicious:

1. Kill the Process:
 - o Right-click the suspicious process and select Kill Process to terminate it immediately.
2. Suspend the Process:
 - o If you are unsure and want to prevent the process from causing damage while you investigate further, right-click and select Suspend.
3. Remove the Source File:
 - o Go to the location of the file (using the Path from the Properties window) and delete the executable file if confirmed to be malware.
 - o Be cautious, as some malware might prevent you from deleting it. In such cases, you may need to boot into Safe Mode or use a specialized tool.

Step 6: Scan Your System

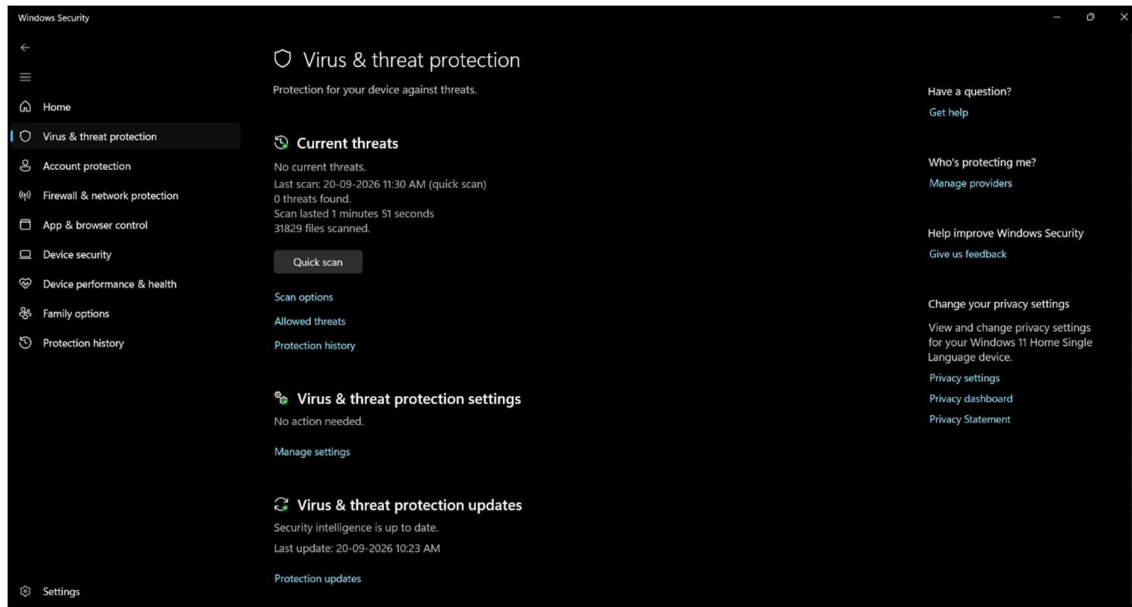
After terminating any suspicious processes, it's a good idea to:

- Run a full antivirus scan using trusted antivirus software.
- Use Malware Removal Tools like Malwarebytes or Windows Defender to perform an in-depth scan of your system.

Example of Identifying a Suspicious Process

1. You notice a process called randomname123.exe using an unusually high amount of CPU.
2. Right-click it, select Properties, and find it is running from a suspicious folder, not the system folder.
3. You check the TCP/IP tab and see it is making external connections to unknown IP addresses.
4. A quick search shows that this process is often associated with a specific type of malware.

5. You kill the process and remove the executable file from the system.



Rubrics:

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:

The Process Explorer tool was successfully used to examine running processes and identify potentially suspicious processes by checking process names, parent-child relationships, digital signatures, file paths, resource usage, company information, and network activity. This practical demonstrated how Process Explorer can assist in detecting and investigating potentially malicious processes in a Windows system.